Pool Ball Detection and Recognition

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Abstract

This report presents a two-stage computer vision pipeline for localising and recognising pool balls. Stage 1 fuses multi-channel CIELAB Canny edges, applies a dynamic pocket and netting mask, and detects circles with Hough, a 40-pixel physical NMS, and a raw-frame recovery path. Stage 2 uses 30/32-bin hue and saturation histograms, a 30/70 saturation-dominant distance, and a split-mask cue gate. Across the six tables the final system achieves Stage 1 precision, recall, and F1 of 0.911, and Stage 2 recognition accuracy of 76.8% on correctly localised balls.

1 Introduction

The task was to localise every pool ball on a rectified table image and classify it as one of the sixteen balls. The six test images contain strong shadows, uneven lighting, dense racks, solids, and stripes. I developed the pipeline through explicit experiments and retained an error analysis of the failures that remain. The repository is at `git@gitlab.scss.tcd.ie:csu33012-2627-group07/project07.git`.

1.1 End-to-End Pipeline Overview

Each image is warped into the 1434×683 metric frame. Stage 1 fuses Canny edges from the $L/a/b$ channels, applies a dynamic pocket and netting mask, and runs Hough Circles with 40-pixel minimum separation, a 22-pixel cushion margin, and 40-pixel NMS. A raw-frame recovery branch runs Standard Canny and Hough on the unwarped image, projects candidates through $\mathbf{H}$, and fuses them when they lie more than 30 pixels from every rectified detection. Stage 2 compares 30/32-bin hue and saturation histograms using a 30/70 distance and a split-mask cue gate, rejecting distances above 0.55 (Figure 1).

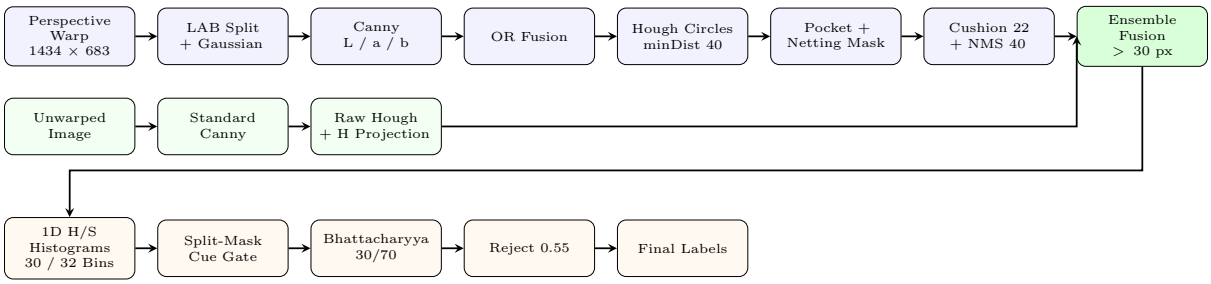


Figure 1: End-to-end pipeline architecture.

2 Stage 1: Ball Detection

2.1 Perspective rectification

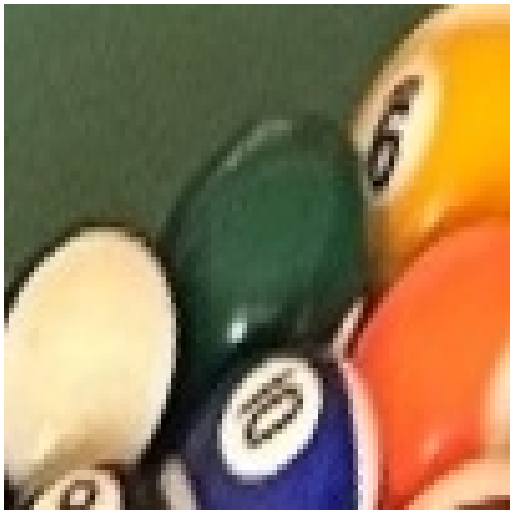
Each photograph is mapped by a planar homography from the four cushion corners into a 1434×683 metric frame matching the $1434\,\text{mm} \times 683\,\text{mm}$ playing area, so one transformed pixel is one millimetre.

2.2 Failed green-channel boost and adaptive thresholding

To recover green balls I first boosted the RGB green channel by 1.5 and applied adaptive Gaussian thresholding (block 15, constant 2). This failed because the green felt was boosted equally and its woven texture dominated the edge map; ball boundaries washed out.

2.3 Failed HSV and single-channel lightness thresholding

I then thresholded HSV with fixed ranges H 35–85, S 40–255, and V 40–255. Green balls 6 and 14 vanished because their hue and saturation overlap the felt. A single-channel lightness threshold also failed: shadows fragmented ball boundaries and dark balls could not be separated from the table. These failures motivated a multi-channel approach.



(a) Region around the green ball.

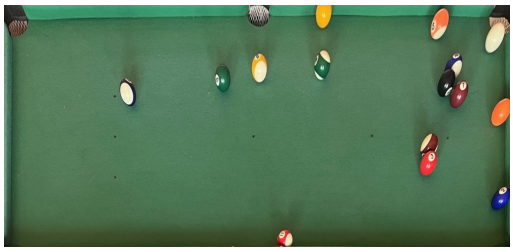


(b) Fixed HSV mask: green ball merges into felt.

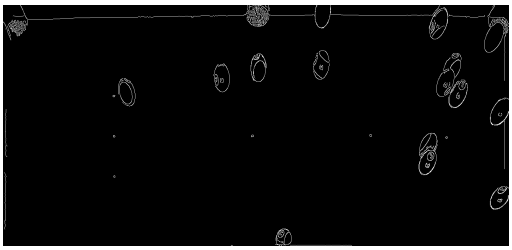
Figure 2: Static HSV thresholding fails for green object balls: the green ball vanishes into the green felt.

2.4 Successful multi-channel CIELAB edge fusion

I split the rectified image into CIELAB, smoothed all channels with a 5×5 Gaussian ($\sigma = 1.2$), and ran Canny independently: thresholds 45/95 on L and 35/85 on a and b. A logical OR fused the three edge maps, capturing both luminance silhouettes and colour boundaries that survive shadow. Figure 3 shows the rectified Table 4 image and its fused edge map.



(a) Rectified Table 4.



(b) Gaussian-smoothed LAB Canny edge map.

Figure 3: Rectification and multi-channel LAB Canny edge fusion.

2.5 Dynamic pocket mask, netting mask, and physical non-maximum suppression

The fused map is passed to Hough with a minimum spatial separation of 40 mm (40 px) and a radius range of 19–24 px. A blind 95-pixel pocket rejection removed genuine corner balls, so I replaced it with a dynamic mask. For each pocket I thresholded a 120×120 region for dark holes ($L < 35$) and bright brackets ($L > 200$), dilated the LAB edges with a 15×15 kernel to capture netting, combined the masks with a logical OR, restricted each to 70 pixels, and dilated the combined mask with 13×13. A candidate is rejected only when its centre lies on this hole–bracket–netting mask. I kept a 22-pixel cushion margin because 5 pixels recovered far-edge balls but reintroduced rail shadows.
Circular Hough voting occasionally returned intersecting peaks for one ball, so I enforce a 40-pixel physical non-maximum suppression. Iterative skeletonization and morphological boundary thinning removed almost nothing (1.025 pixels per skeleton pixel), and an accumulator-threshold sweep made ghosts disappear only at a threshold of 14 while true balls fell from 83 to 15. I therefore added a raw-frame path: gradient-based edge detection and circular Hough voting on the unwarped image (accumulator threshold 14, minimum spatial separation 30 px, radius 18–30 px), projected through $\mathbf{H}$, and fused only if more than 30 pixels from every rectified detection. This recovered three balls, raising TP from 79 to 82 and F1 from 0.893 to 0.911, with FP unchanged.

3 Stage 2: Ball Recognition and Classification

3.1 Failed K-Means clustering

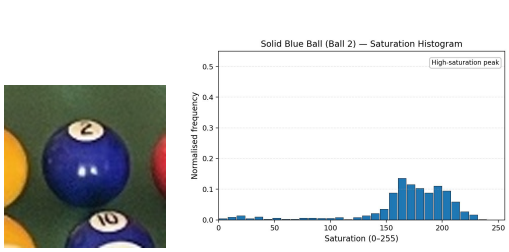
I first tried $K = 2$ K-Means on each ball mask and compared cluster colours with the references. It was slow and rotation-dependent: on a solid ball the clusters split the number and specular highlights rather than producing a stable stripe descriptor.

3.2 Successful saturation-dominant histograms

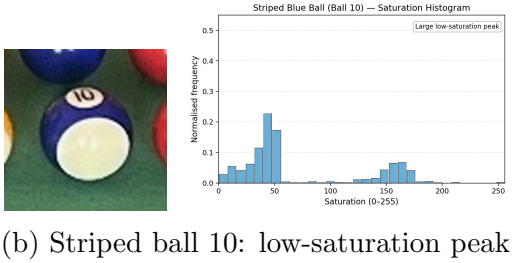
I switched to 1D histograms: 30 hue bins and 32 saturation bins from a 16-pixel circular mask, compared with the sixteen references by Bhattacharyya distance. A white stripe lowers saturation regardless of rotation, so I weight hue 0.30 and saturation 0.70 and reject distances above 0.55:

$$d = 0.30\, d_{\text{hue}} + 0.70\, d_{\text{sat}}.$$

Tightening to 0.55 removed circular rail shadows. The cue histogram overlapped green-felt reflections, so I added a split-mask gate: a 16-pixel centre for brightness and whiteness and a 21-pixel mask for overall paint. Centre value > 235, centre saturation < 75, and full saturation < 76 excluded the purple 12 and 9 while retaining the cue; the Highlander rule selects the lowest full saturation.
Figure 4 compares saturation histograms of solid ball 2 and striped ball 10. The solid ball peaks at high saturation; the white stripe creates a distinct low-saturation peak independent of rotation.



(a) Solid ball 2: high-saturation peak.



(b) Striped ball 10: low-saturation peak from the white stripe.

Figure 4: Blue-ball comparison: each crop sits beside its 1D saturation histogram.

4 Results and Error Analysis

4.1 Stage 1: Detection Results

Stage 1 uses one-to-one greedy matching within 21.5 pixels, half the 43-pixel diameter. A detected centre is a True Positive when its distance to an unmatched ground-truth centre satisfies

$$\sqrt{(x_{\text{det}} - x_{\text{gt}})^2 + (y_{\text{det}} - y_{\text{gt}})^2} \leq 21.5 \text{ px}.$$

An unmatched detection is a False Positive and an unmatched ground-truth centre is a False Negative. Precision, recall, and F1 are

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

Table 1 reports per-table and overall metrics; overall precision, recall, and F1 are all 0.911.

Table 1: Stage 1 detection metrics after dynamic pocket masking.

Image	TP	FP	FN	Precision	Recall	F1-score
Table 1	13	0	3	1.000	0.812	0.897
Table 2	15	2	1	0.882	0.938	0.909
Table 3	16	0	0	1.000	1.000	1.000
Table 4	14	2	1	0.875	0.933	0.903
Table 5	13	2	1	0.867	0.929	0.897
Table 6	11	2	2	0.846	0.846	0.846
Overall	82	8	8	0.911	0.911	0.911

4.2 Stage 2: Recognition Results

Stage 2 accuracy is measured only over Stage 1 True Positives:

$$\text{Recognition Accuracy} = \frac{\text{correctly labelled Stage 1 TPs}}{\text{total Stage 1 TPs}}.$$

A TP is wrong when its 30/70 weighted distance is closest to another reference or exceeds 0.55. Across six tables, 63 of 82 correctly localised balls receive the correct label (76.8%).

Table 2: Stage 2 recognition accuracy on correctly localised balls.

Image	Correct labels	True positives	Accuracy
Table 1	10	13	0.769
Table 2	10	15	0.667
Table 3	12	16	0.750
Table 4	13	14	0.929
Table 5	10	13	0.769
Table 6	8	11	0.727
Overall	63	82	0.768

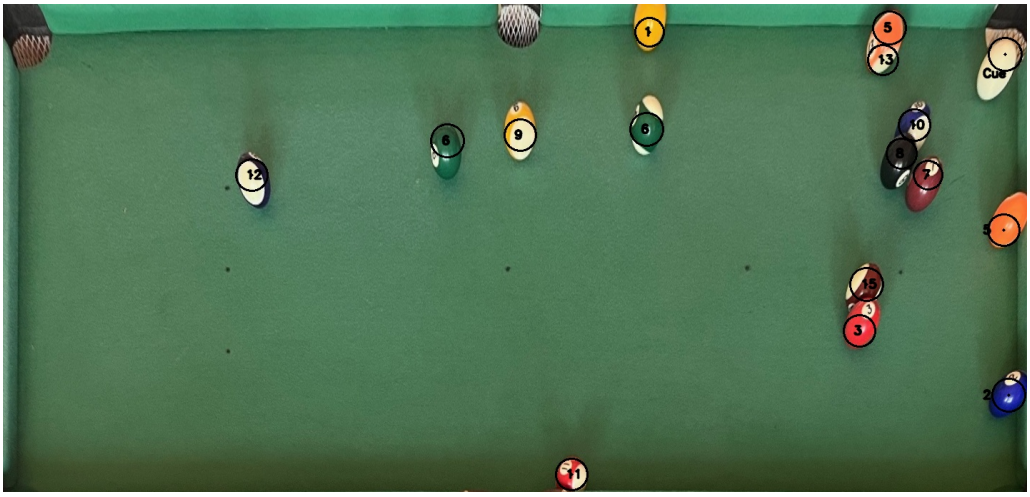


Figure 5: Final labelled output for Table 4; Stage 2 recognition accuracy is 92.9%.

Figure 5 shows the Table 4 output, which combines high Stage 1 recall with 92.9% Stage 2 accuracy.

4.3 Error Analysis

After the 22-pixel cushion margin, split-mask cue gate, and raw-frame fusion, Stage 1 yields 82 TP, 8 FP, and 8 FN; Stage 2 reaches 76.8%. I audited Table 2 because it isolates the active failures: duplicate perimeter detections, missed interior balls in the dense cluster, and rotation/specular misclassification. The cue and 9 are resolved. Figure 6 shows the overlay.

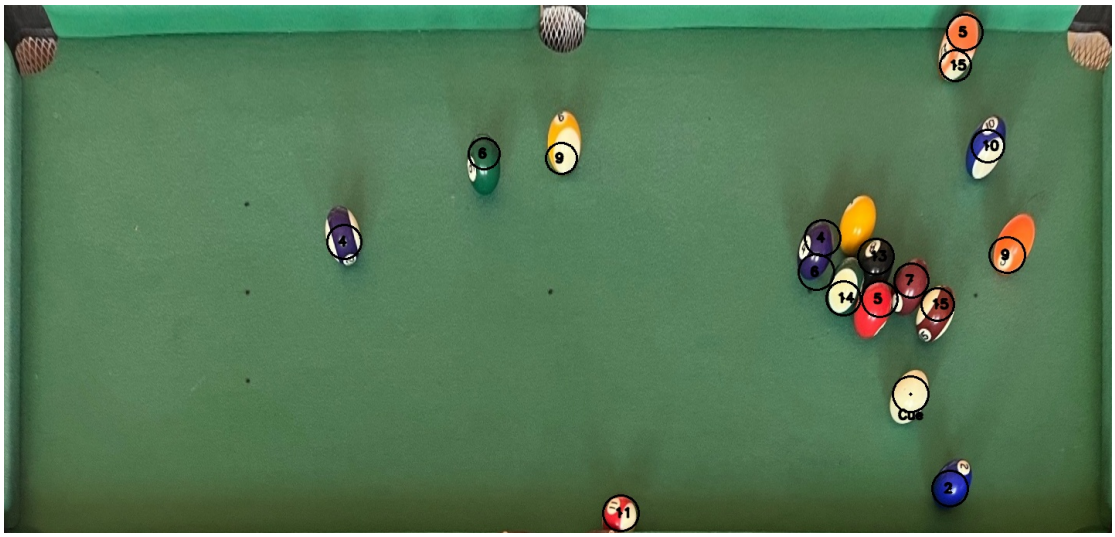


Figure 6: Final labelled output for Table 2, illustrating duplicate detections on single perimeter balls (top right) and missed detections inside dense clusters.

Failure Mode 1: Duplicate Detections on Single Balls. In my evaluation I observed that candidates near a table boundary can cast two accumulator peaks for the same physical ball. The cushion truncates the perimeter arc, and the two surviving arcs on opposite sides can produce centres more than 40 pixels apart, so they lie just outside the 40-pixel physical NMS and both survive. Figure 6 shows this at the top right of Table 2, where one ball receives two overlapping circles labelled 5 and 15. One is correct and the other is an artificial false positive; only a duplicated circular response is involved.

Failure Mode 2: Dense-Cluster Gradient Nullification. In the right-hand cluster of Table 2 I observed that touching balls break the continuous Canny arcs Hough needs. Adjacent boundaries merge into one irregular gradient, so accumulator votes spread and the interior peak stays below the Hough accumulator threshold of 8.0. Lowering the threshold admits shadow arcs, so I kept the stricter setting. This is the remaining false negative: the ball is visible but has no isolated circular evidence. A Watershed pass before circle fitting is the natural remedy.

Failure Mode 3: Rotation and Specular Misclassification. My evaluation showed that rotation and specular reflections shift 1D hue/saturation bins even when the centre is correct. A white number circle or highlight can dominate the saturation peak, while shadow can suppress the paint peak; the 30/70 distance then lands closest to the wrong reference or exceeds 0.55. Table 2 exposes this in its 66.7% Stage 2 accuracy, where balls 3, 8, 12, and 13 receive neighbouring labels or are pulled toward the cue reference. I kept the deterministic classifier because the errors are sparse; a specular-rejection mask or rotation-invariant stripe descriptor is the next refinement.

5 Discussion

Final Stage 1 P/R/F1 is 0.911/0.911/0.911; Stage 2 is 0.768. LAB fusion was decisive: L gave the silhouette while a/b recovered colour boundaries in shadow; the cost was a

denser edge map, which made the 40-pixel NMS and 22-pixel cushion margin necessary. The raw-frame path added three TPs without FPs, but lowering the Hough threshold alone traded precision for recall. For recognition, saturation histograms beat K-Means; the 30/70 weighting encodes the stripe effect, and the split-mask gate prevents a bullseye 9 or 14 from hijacking the cue. Remaining limits are duplicate perimeter responses, dense-cluster merging, and specular/stripe overlap; a spatial stripe descriptor and Watershed segmentation are the next steps.

6 Conclusion

I implemented a two-stage pipeline that rectifies the table, fuses LAB Canny edges, filters Hough candidates with a dynamic pocket mask and physical NMS, recovers perimeter balls through a raw-frame path, and classifies survivors with saturation-dominant histograms and a split-mask cue gate. It achieves Stage 1 P/R/F1 of 0.911 and Stage 2 accuracy of 76.8%. The main residual errors are duplicate perimeter detections, dense-cluster misses, and specular/stripe confusion; Watershed segmentation and a spatial stripe descriptor are the natural next steps.

AI Usage Statement

Generative AI tools (Google Gemini) were used for report text revisions, LaTeX/TikZ formatting, and standalone C++ diagnostic scripts for parameter sweeps. I implemented all algorithms, selected parameters, ran experiments, and verified the reported results.

Prompt & Response Example 1 (Parameter Sweep Script): *User Prompt:* "Write a diagnostic script to sweep the Hough accumulator threshold from 6 to 18 and compare edge-thinned LAB maps with raw Canny maps." *AI Response:* "Generated a C++ benchmark logging TP/FP counts to stage2_feature_analysis.csv."

Prompt & Response Example 2 (Gate Suggestion): *User Prompt:* "The 1D histogram labels the 9-ball as the cue when its white circle faces the camera. How can I separate them?" *AI Response:* "Suggested a split-mask saturation/value check comparing the 16px centre with the 21px full ball."

Prompt & Response Example 3 (LaTeX Troubleshooting): *User Prompt:* "The pipeline TikZ block wraps awkwardly and overflows the margin. How do I format a two-stage flow?" *AI Response:* "Provided matrix/node code with explicit node distances and rounded styles."

All core algorithmic implementations, experimental designs, parameter selections, and final data verification were conducted directly by the author.